PROBLEM 21

Apparent Weight on a Spinning Planet

Original statement · Problems.pdf, p. 10

Centrifugal Force

On a spinning spherical planet of radius R and gravitational acceleration g_p, the bodies weight [sic] twice as much on the poles as on the equator. Determine the duration of the day on that planet.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a spherical planet of radius R spins with constant angular velocity Ω about its polar axis. At its surface the gravitational attraction is g_p per unit mass, directed toward the centre. A body of mass m rests on the ground.
- **Why the planet frame.** "Weight" is what a scale fixed to the ground reads: it is defined in the frame of the planet, where the body is at rest. In that frame the dynamics reduce to a static balance once the inertia forces (Notes Eq. 4.62) are added; the centrifugal force is what makes the weight depend on latitude.
- **Hypotheses:**
  1. The frame S_0 with origin at the centre O of the planet and axes fixed with respect to the stars is inertial (O unaccelerated).
  2. Ω is constant; the planet is a rigid sphere and g_p is the same everywhere on its surface (spherical mass distribution).
  3. The apparent weight is the magnitude of the contact force $\mathbf{N}$ the ground (or a scale) exerts on the body at rest.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $\mathbf{k}_0$ along the polar axis (towards the north pole).
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the planet, $\mathbf{k}_1 = \mathbf{k}_0$, with $\mathbf{i}_1$ in the equatorial plane and in the meridian plane of the body; S_1 coincides with S_0 at $t = 0$.

- **Radial unit vector:** $\mathbf{e}_r$, from O to the body; deduced in the next section.
- **Coordinates and degrees of freedom:** the body is at rest in S_1 at latitude λ (angle of OP above the equatorial plane): there is no motion to determine; the unknown is Ω (and then the day $T_{\text{day}} = 2\pi/\Omega$).

Symbol	Meaning	SI unit
R	radius of the planet	m
g_p	gravitational acceleration at the surface	m/s^2
Ω	spin rate of the planet	rad/s
λ	latitude of the body	rad
m	mass of the body	kg
T_{day}	duration of the day (one rotation, sidereal)	s

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from the north pole

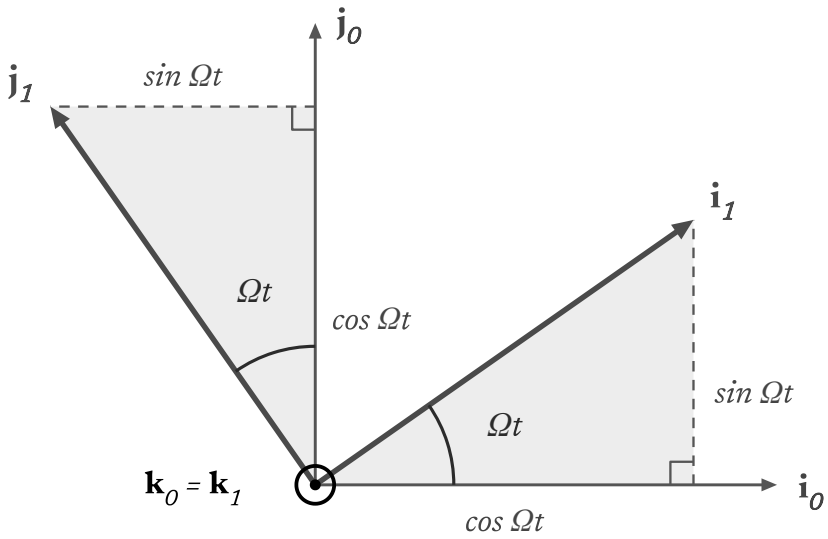


Figure 1. View from above the north pole ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). $\mathbf{i}_1$ is $\mathbf{i}_0$ turned counter-clockwise by Ωt (legs $\cos \Omega t$, $\sin \Omega t$); $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, so its opposite leg points along $-\mathbf{i}_0$.

The planet turns about $\mathbf{k}_0$, so the equatorial plane is drawn looking down from $+\mathbf{k}_0$ (from above the north pole): $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, by the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$, $\mathbf{j}_0$ up the page. With $\Omega > 0$ the planet turns counter-clockwise, and after a time t the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is Ωt:

- $\mathbf{i}_1$: hypotenuse 1, adjacent leg $\cos \Omega t$ along $\mathbf{i}_0$, opposite leg $\sin \Omega t$ along $+\mathbf{j}_0$, so $\mathbf{i}_1 = \cos \Omega t\, \mathbf{i}_0 + \sin \Omega t\, \mathbf{j}_0$.
- $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, towards the left: adjacent leg $\cos \Omega t$ along $\mathbf{j}_0$, opposite leg $\sin \Omega t$ along $-\mathbf{i}_0$, so $\mathbf{j}_1 = -\sin \Omega t\, \mathbf{i}_0 + \cos \Omega t\, \mathbf{j}_0$ (the minus sign).
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t = \cos^2\Omega t + \sin^2\Omega t = 1$$

The columns have unit length and $-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. Poisson check (chain rule): $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega\sin\Omega t\,\mathbf{i}_0 + \Omega\cos\Omega t\,\mathbf{j}_0 = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The position of the body in its meridian plane

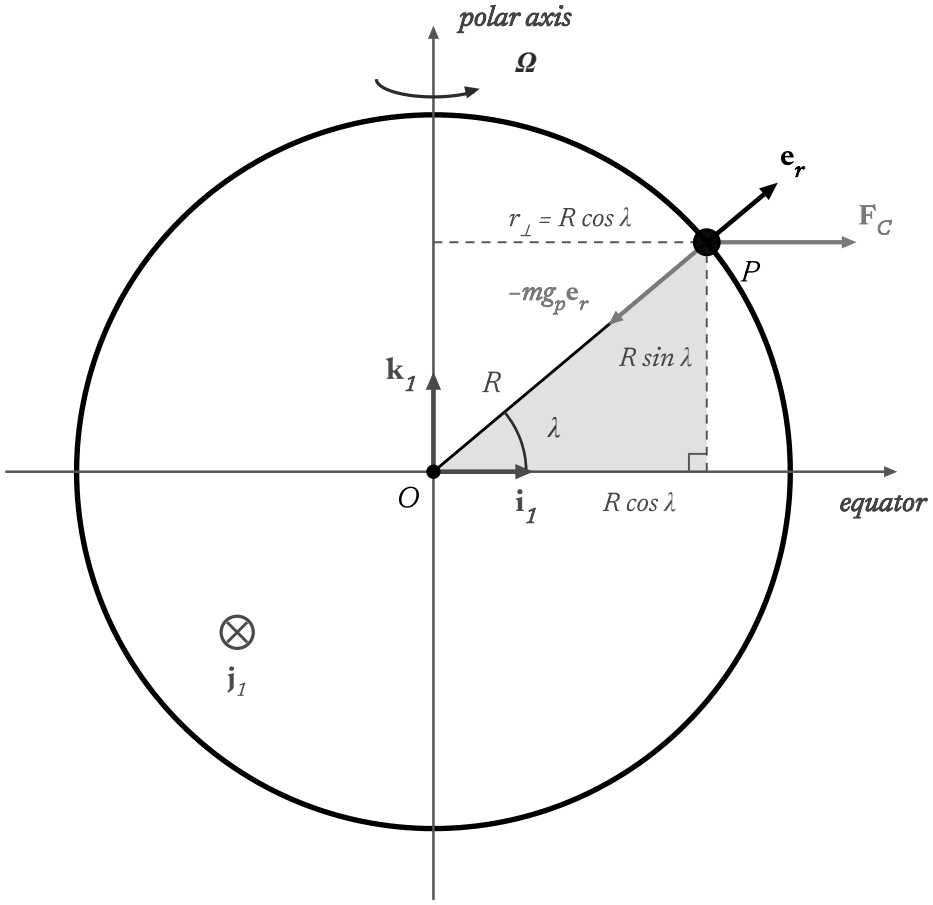


Figure 2. Meridian plane of the body ($\mathbf{i}_1$ right, $\mathbf{k}_1$ up, $\mathbf{j}_1$ into the paper, $\otimes$). In the shaded right triangle OFP the hypotenuse is R, the leg adjacent to λ is $R\cos\lambda$ along $\mathbf{i}_1$ and the opposite leg is $R\sin\lambda$ along $\mathbf{k}_1$. The centrifugal force $m\Omega^2\mathbf{r}_\perp$ is horizontal, away from the axis; the attraction $-mg_p\mathbf{e}_r$ points to O.

The meridian plane of the body contains $\mathbf{i}_1$ and $\mathbf{k}_1$; it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up, so $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points into the paper ($\otimes$; fingers from "up" to "right", clockwise, thumb away from the reader).

P is on the surface, at distance R from the centre: $\mathbf{r}_1^P = \overrightarrow{OP} = R\,\mathbf{e}_r$. Let F be the foot of the perpendicular from P to the equatorial line. In the shaded right triangle OFP (right angle at F) the hypotenuse is $|OP| = R$, the leg adjacent to λ is $OF = R\cos\lambda$ along $\mathbf{i}_1$ and the leg opposite to λ is $FP = R\sin\lambda$ along $\mathbf{k}_1$:

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{e}_r = \cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1$$

(here the latitude is the usual polar angle of the meridian plane, measured from $\mathbf{i}_1$ towards $\mathbf{k}_1$, so no shift is needed). The part of $\mathbf{r}_1^P$ normal to the axis is the horizontal dashed segment from the axis to P, of length equal to the leg OF:

$$\mathbf{r}_\perp = R\cos\lambda\,\mathbf{i}_1$$

Checks: $\lambda = \pi/2$ gives $R\,\mathbf{k}_1$ (north pole, $\mathbf{r}_\perp = \mathbf{0}$); $\lambda = 0$ gives $R\,\mathbf{i}_1$ (equator, $r_\perp = R$).

Kinematics

The body is at rest in S_1: λ and R are constant and $\mathbf{i}_1, \mathbf{k}_1$ are fixed in S_1, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$: there are no relative-motion terms and no Coriolis force. (Its absolute acceleration, by Notes Eq. 3.27, is only the centripetal term $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = -\Omega^2 R\cos\lambda\,\mathbf{i}_1$.)

Forces and Equations of Motion

1. Forces on the body in the planet frame

Pedagogical Justification: the body is at rest in S_1, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In Notes Eq. 4.63, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, the Coriolis force vanishes ($\mathbf{v}_1^P = \mathbf{0}$), $\mathbf{F}_A = \mathbf{0}$ and $\mathbf{F}_E = \mathbf{0}$; only the centrifugal force remains.

Actual forces: attraction $-mg_p\,\mathbf{e}_r$ and contact force $\mathbf{N}$. Centrifugal force (Slide 30), with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\mathbf{F}_C = -m\,\Omega\mathbf{k}_1 \times (\Omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\Omega\mathbf{k}_1 \times (R\Omega\cos\lambda\,\mathbf{j}_1) = m\Omega^2 R\cos\lambda\,\mathbf{i}_1 = m\Omega^2\,\mathbf{r}_\perp$$

It is horizontal and points away from the axis, as drawn.

Static balance in S_1

The equation of motion is the static balance of Notes Eq. 4.63 with $\mathbf{a}_1^P = \mathbf{0}$:

$$\mathbf{0} = -mg_p\,\mathbf{e}_r + \mathbf{N} + m\Omega^2 R\cos\lambda\,\mathbf{i}_1 \quad\Longrightarrow\quad \mathbf{N} = mg_p\,(\cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1) - m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

Answers and Checks

2. Weight at the pole and at the equator

- **Pole** ($\lambda = \pi/2$): $\cos\lambda = 0$, $\mathbf{r}_\perp = \mathbf{0}$, no centrifugal force:

$$\mathbf{N}_{\text{pole}} = mg_p\,\mathbf{k}_1, \qquad W_{\text{pole}} = mg_p$$

- **Equator** ($\lambda = 0$): $\mathbf{e}_r = \mathbf{i}_1$ and the centrifugal force is radial and outward:

$$\mathbf{N}_{\text{eq}} = m\left(g_p - \Omega^2 R\right)\mathbf{i}_1, \qquad W_{\text{eq}} = m\left(g_p - \Omega^2 R\right)$$

A body can rest on the equator only if $N_{\text{eq}} \geq 0$ (the ground can push but not pull), i.e. $\Omega^2 R \leq g_p$.

3. Condition $W_{\text{pole}} = 2W_{\text{eq}}$ and length of the day

$$mg_p = 2m\left(g_p - \Omega^2 R\right) \quad\Longrightarrow\quad 2\Omega^2 R = 2g_p - g_p = g_p \quad\Longrightarrow\quad \Omega = \sqrt{\frac{g_p}{2R}}$$

(this satisfies $\Omega^2 R = g_p/2 \leq g_p$. The other possibility, $mg_p = 2m(\Omega^2 R - g_p)$, would mean $\Omega^2 R > g_p$: equatorial bodies would fly off, so it is rejected.) The day is the time of one rotation:

$$\boxed{T_{\text{day}} = \frac{2\pi}{\Omega} = 2\pi\sqrt{\frac{2R}{g_p}}}$$

Checks

- **Meaning.** The centrifugal force at the equator cancels half of the attraction, so $\Omega^2 R = g_p/2$: the planet spins at $1/\sqrt{2} \approx 0.71$ of the break-up rate $\sqrt{g_p/R}$ at which equatorial bodies would float ($W_{\text{eq}} = 0$).
- **Any latitude** (extension, same balance): $|\mathbf{N}|^2 = m^2\left[g_p^2 - \left(2g_p\Omega^2 R - \Omega^4 R^2\right)\cos^2\lambda\right]$, which with $\Omega^2 R = g_p/2$ gives $W(\lambda) = mg_p\sqrt{1 - \tfrac{3}{4}\cos^2\lambda}$: from $mg_p/2$ at the equator to mg_p at the poles (e.g. $mg_p\sqrt{10}/4 \approx 0.79\,mg_p$ at 45°). Away from the poles and the equator $\mathbf{N}$ is not radial: the plumb line does not point to the centre.
- **Numbers** (illustration only, Earth-like $R = 6.371 \times 10^6$ m, $g_p = 9.81$ m/s^2): $T_{\text{day}} \approx 7.16 \times 10^3$ s ≈ 1.99 h.
- **Limit:** a slower spin ($\Omega \to 0$) gives $W_{\text{eq}} \to W_{\text{pole}}$; the ratio 2 requires a fast rotation.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{j}_1 = 0$ and $|\mathbf{e}_r| = \sqrt{\cos^2\lambda + \sin^2\lambda} = 1$.
- **Dimensions:** $[\Omega^2 R] = \mathrm{s}^{-2}\,\mathrm{m} = [g_p]$; $[T_{\text{day}}] = \sqrt{\mathrm{m}/(\mathrm{m\,s^{-2}})} = \mathrm{s}$.
- **Verification (sympy):** the inertial acceleration of a point at rest on the planet, obtained by differentiating $[{}_0R_1]\{\mathbf{r}_1^P\}$ twice, equals $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$; the balance gives $W_{\text{pole}} = mg_p$, $W_{\text{eq}} = m|g_p - \Omega^2 R|$, $\Omega = \sqrt{g_p/(2R)}$ and $T_{\text{day}} = 2\pi\sqrt{2R/g_p}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 21 in its solution keys, so the results are verified by independent derivation (the sympy check above).

Source: Problems.pdf (PDF page 10); no official key